

KEYSTONE SUBSURFACE INSPECTION

DISTRIBUTION MAIN CONDITION ASSESSMENT

Report reference: CA-2025-0279

Date of inspection: 2025-02-24

Inspector: M. Okonkwo, P.E.

Client: Three Rivers Water Authority

1. SCOPE

Visual and limited non-destructive assessment of distribution main segment WM-1378, located on Butler Street in the Lawrenceville area of the PZ-HL1 pressure zone. The segment is a 16-inch cast iron main of approximately 394 ft, originally installed in 1902.

2. METHOD

Access was obtained by test pit at one location. The exposed section was cleaned and examined visually. Wall thickness was assessed by ultrasonic spot readings where surface condition permitted. No internal inspection was performed.

3. OBSERVATIONS

Extensive external corrosion with deep pitting and confirmed graphitization over an estimated 40% of the exposed circumference. Remaining wall thickness is materially reduced.

No active leak observed. Soil at pipe depth was damp with poor drainage.

4. FINDING

Condition classification: SEVERE

Primary observation: external corrosion

5. RECOMMENDATION

Prioritise this segment for replacement planning. Re-inspect within 12 months if replacement cannot be scheduled.

This assessment reflects the condition of the exposed section only and should not be taken as representative of the entire segment.